Overview. My research interests lie in networking and hardware/software co-design for datacenter systems. Through more than four years of work on offloading host functions to programmable hardware, I have explored how to divide responsibilities between hardware data planes and software control planes. Growing memory demands from KV caches in large language models have led me to a research question: *What can programmable NICs contribute to memory pooling for LLM inference* ***beyond what software alone can achieve****?* During my Ph.D., I hope to begin with CXL-expanded memory servers that provide a KV-cache offloading tier for GPU-equipped compute nodes.

Research Experiences. During my M.S. at UNIST, I worked with **Prof. Youngbin Im** on FlexServer[1], a research project focused on optimizing content servers that deliver files from storage to clients over the network. Reading files from storage can stall the server because the process must wait until the file data has been loaded into the TCP send buffer before continuing with other work. To address this, I implemented a transmission system built on multi-tier caching that prefetches hot files from NVMe SSDs into host DRAM and SmartNIC memory. I implemented this system on both BlueField and Agilio SmartNICs. A key challenge was coordinating the NIC's data-transfer capabilities with software buffer management while preserving correct TCP behavior and a zero-copy design. I reorganized the TCP send buffer to handle data according to its location: requesting the SmartNIC to transmit data stored in its memory, passing physical addresses to the NIC for data in host memory, and using an optimized I/O engine to read data from NVMe SSDs into the TCP send buffer. FlexServer achieved up to 351 Gbps, outperforming the state-of-the-art baseline, IO-TCP[2], by 4.4×. I am the first author of the resulting paper, which is available as a preprint. However, the pipeline still used a bounce buffer to stage data from NVMe devices before network transmission. This limitation led me to explore the hardware capabilities and software support needed for direct data transfers between NVMe devices and FPGA-based SmartNICs.

At MangoBoost, I worked on software stacks for NVMe/TCP accelerators built on FPGA-based SmartNICs. As a coauthor of BoostX-NTI[3], accepted at ISCA 2026, I developed its PCIe interface and NVMe-compliant control plane. Since managing complex state in hardware is challenging, we ran the control plane on an onboard Arm processor to handle NVMe register configuration and admin commands, while keeping performance-critical NVMe I/O processing in FPGA logic. A key challenge was coordinating these components to preserve correct NVMe behavior during network faults, for which I developed fault-handling mechanisms. One example is *multipath*, which allows NVMe I/O to continue over an alternative network path when one path becomes unavailable. The resulting accelerator exposed remote NVMe/TCP storage as a local NVMe/PCIe device, supported Windows and Linux hosts, and achieved UNH-IOL certification[4]. By executing the entire I/O path in hardware, it sustained line-rate throughput over commodity Ethernet, matching or exceeding NVMe/RDMA (up to 1.24× for 4 KiB I/O). This project taught me that only a simple, stable data path is worth hardening in hardware, while stateful control such as fault recovery belongs in software.

Motivation. My work on FlexServer and BoostX-NTI showed me how NIC features can improve system efficiency by offloading computation and reducing unnecessary data movement. More recently, my team at MangoBoost has been developing a storage solution for KV-cache offloading. I evaluated our solution using Mooncake[5], which led me to consider DRAM-based KV-cache sharing and whether a dedicated server providing memory to other nodes could help with KV-cache offloading. While my current work there focuses on software-based solutions, for my Ph.D. I want to explore how programmable NICs can support KV-cache sharing across memory servers. As LLM services sustain more concurrent multi-turn conversations with longer contexts, KV-cache capacity requirements may grow beyond what each server's DIMM slots can cost-effectively accommodate. NVIDIA's BlueField-4-based CMX platform[6] addresses growing context-storage demands with an Ethernet-attached flash tier, illustrating a role for programmable hardware in KV-cache delivery. I am interested in whether CXL-expanded memory servers[7] could provide an intermediate KV-cache

tier between GPU nodes' host DRAM and flash-based context storage such as CMX, with programmable NICs managing transfers to and from GPU nodes[8].

KV Cache Offloading to CXL Memory. Meta's Vistara MemServer[9] demonstrates how CXL can expand memory capacity beyond host DIMM-slot limits and reuse older-generation DRAM in production ML parameter servers. Prior work[9–12] has explored page placement, memory-access monitoring, and resource management for CPU workloads, techniques that can be beneficial for workloads with skewed request patterns. A KV-cache memory server, however, could hold the contexts of many active multi-turn conversations awaiting their next turn. These contexts form a working set that may be read again soon, rather than simply cold data offloaded for long-term storage. When many conversations resume at similar rates, much of this working set may be similarly hot, leaving less access skew for hotness-based placement to exploit. In this setting, aggregate transfer bandwidth may matter more than distinguishing hot from cold blocks. Transfers from host DRAM are constrained by memory-channel bandwidth, while transfers from CXL memory are additionally constrained by the CXL link over PCIe. Distributing KV-cache blocks across the two regions in proportion to their effective bandwidths may improve aggregate transfer throughput and help avoid GPU stalls. However, fully exploiting CXL bandwidth may push CXL memory toward its bandwidth limit, where its access latency can become higher and less stable than local DRAM[13], so if requests to both regions share an ordered NIC queue, slower CXL accesses may cause head-of-line blocking. Since GPU nodes would read KV caches with one-sided RDMA, bypassing the memory server's host software, I want to explore whether coordination between programmable NICs and CXL memory controllers, such as signals that let the NIC pace CXL-bound requests, can prevent such blocking.

In the longer term, I want to extend memory pooling to rack scale and jointly optimize the GPU-equipped compute nodes that access the pool. I hope to explore how RDMA and CXL fabric topologies affect memory sharing and KV-cache delivery at this scale.

Why UW. I would like to work with **Prof. Simon Peter**. NEMO's programmable memory-controller telemetry engine[11] could be extended to support coordination between programmable NICs and CXL memory controllers, which I hope to explore. His use of *pipeline parallelism* in FlexTOE[14] and LineFS[15] shaped my approach to low-level systems: *a request begins at a queue pair, traverses a processing pipeline, and completes at a queue pair.* This perspective was helpful while I was working on BoostX-NTI. I would also like to work with **Prof. Arvind Krishnamurthy**. I am interested in exploring how BlueField's PCIe device emulation, used in SG-IOV[16], could contribute to the design of rack-scale systems. SG-IOV's use of PCIe device emulation connects directly to my work on BoostX-NTI, where I implemented the interface and control plane that exposed remote NVMe/TCP storage as a local NVMe/PCIe device.

My long-term goal is to lead research that enables the effective use of programmable hardware in datacenter systems, whether in academia or at an industrial research lab. Both paths require defining problems from first principles rather than following product roadmaps, and I hope to develop that ability through a Ph.D.

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